THE PENTACLE: A WIRELESS INTERFACE FOR NEW MUSICAL EXPRESSION

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ABSTRACT

This paper describes a battery powered wireless interface focuses on free spatial movement for live musical interaction. The interface is a dodecahedron housing a specially designed 'brain' consisting of an Arduino connected to a direction-sensing sensor-fusion device offering nine degrees of freedom, twelve independently responsive contact microphones and twelve multi-colour leds. The outer surface of the dodecahedron serves as a functional physical input that allows the user to strike or press each of the faces whilst all directional movement of the device is simultaneously transmitted. Messages are converted to OpenSound Control("OSC") from a wireless receiver, and both hardware and software projects are developed continuing an inheritance of open-source licensed systems.

Introduction

The interface engages with issues of how to bridge the divide between physical and digital environments by taking established ideas about Human Computer Interaction (HCI) and extending possibilities into a three dimensional musical interaction. The central hardware of the device is two open source projects, the Funnel IO[1] (aka Arduino FIO) and FreeIMU[2]. The FIO is a device for the hardware realization of the Funnel Server tool-kit, developing a platform and libraries for interacting with hardware in any number of programming-languages and scenarios[3]. The FreeIMU is a hardware project developing decentralized sensor fusion technology and software libraries. The vibration sensing microphone inputs on each face further emphasize the physical nature of input on the device.

The core performance concept is that one can be presented with a wireless object and every gesture made will produce a "different but repeatable,"[4] outcome, with visual indicators that one can track without reference to a

For musical applications, the shaping of the ball was chosen to illustrate the applicability of geometric concepts in musical expression. The device considers the translation between forces, with vibration and movement as an input and light and sound as the output.

The casing and mounting are all made from a folded piece of cardboard. This avoids the common difficulties of realizing workable control surfaces and casings for interfaces and coexists with the ethos of an open source hardware and software solution that the user can experiment with and modify at all levels of technology development.

The core mission of this interface is to allow live performance with digital media to be more tangible, involving and accessible. Actions of movement and spatial gesture become sources of free, unimpeded, accurate information that acts as bridge between physical and digital environments.

References

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